

$$= 2.36 \text{ N}$$

A1

2 (a) $g = (6.1 \pm 0.1) \text{ N m}^{-1}$
 Force = $mg = 6.1 \times 20000$
 $= (122\,000 \pm 2000) \text{ N}$

C1

A1

(b) $F = \frac{mv^2}{r}$ or $g = \frac{v^2}{r}$

C1

$$v = \sqrt{(6.1 \times (8.2 \times 10^6))}$$

$$= (7.1 \pm 0.1) \times 10^3 \text{ m s}^{-1}$$

A1

(c) (i) The gravitational potential at a point is defined as the work done per unit mass in bringing a small test mass from infinity to that point.

B1

(ii) $\phi = -\frac{GM}{r} = -gr$
 $= -(4.0 \pm 0.1) \times 10^7 \text{ J kg}^{-1}$

C1

A1

OR

recognizes that this is the area under the graph from point to infinity

B1

counting squares gives total area = $-(4.0 \pm 2.0) \times 10^7 \text{ J kg}^{-1}$

B1

3 (a) (i) a and r are constant, so a is proportional to x

B1